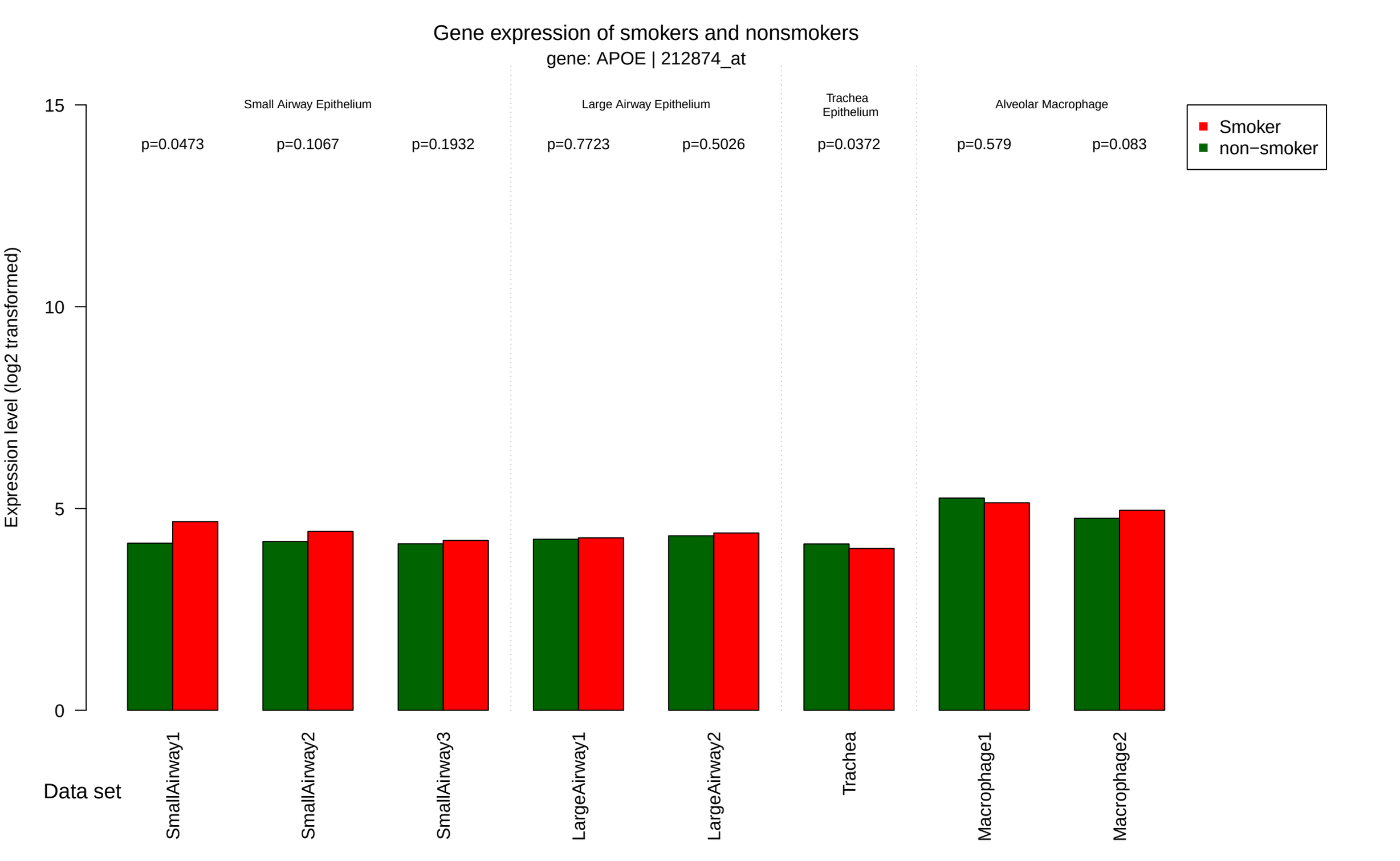
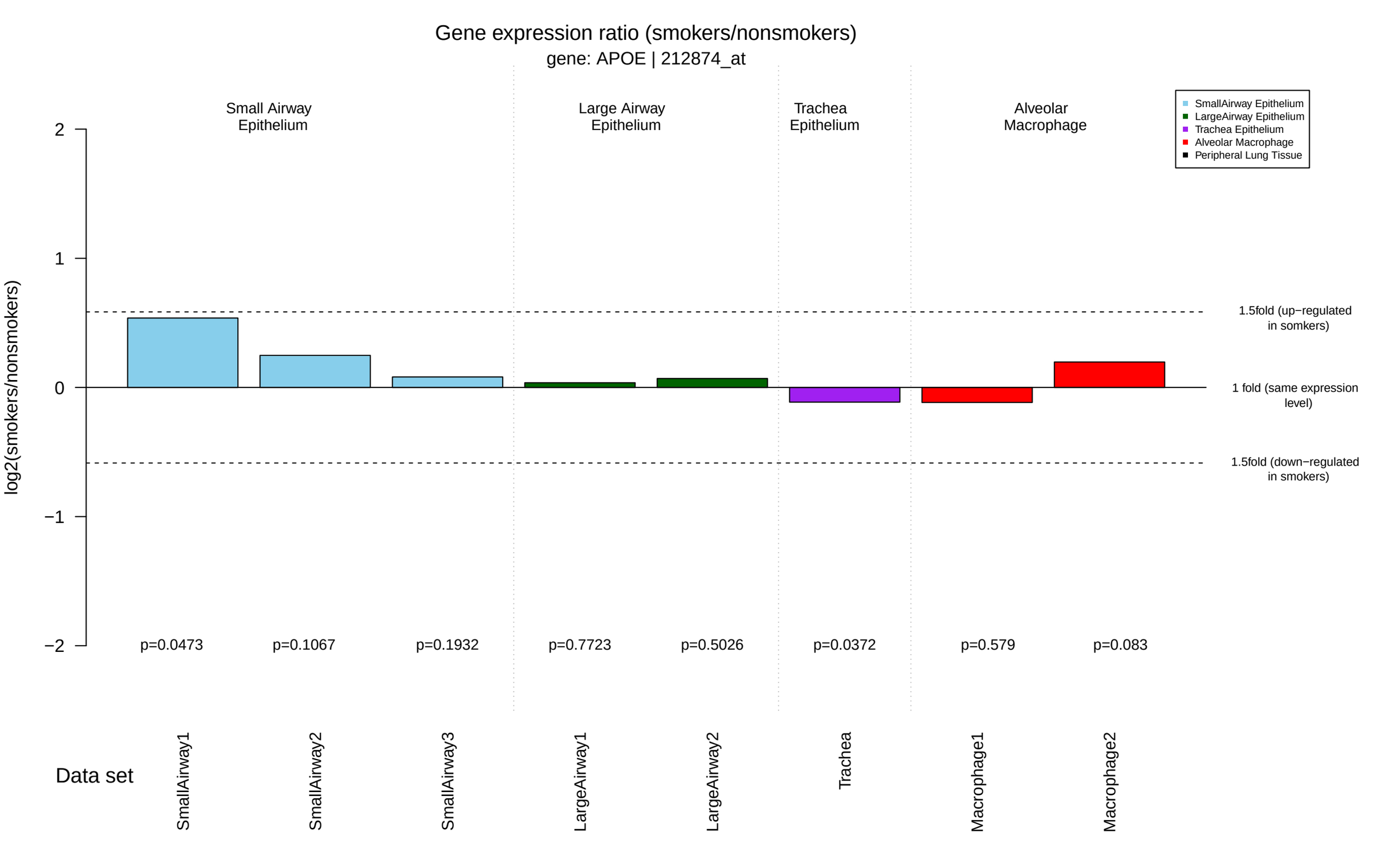
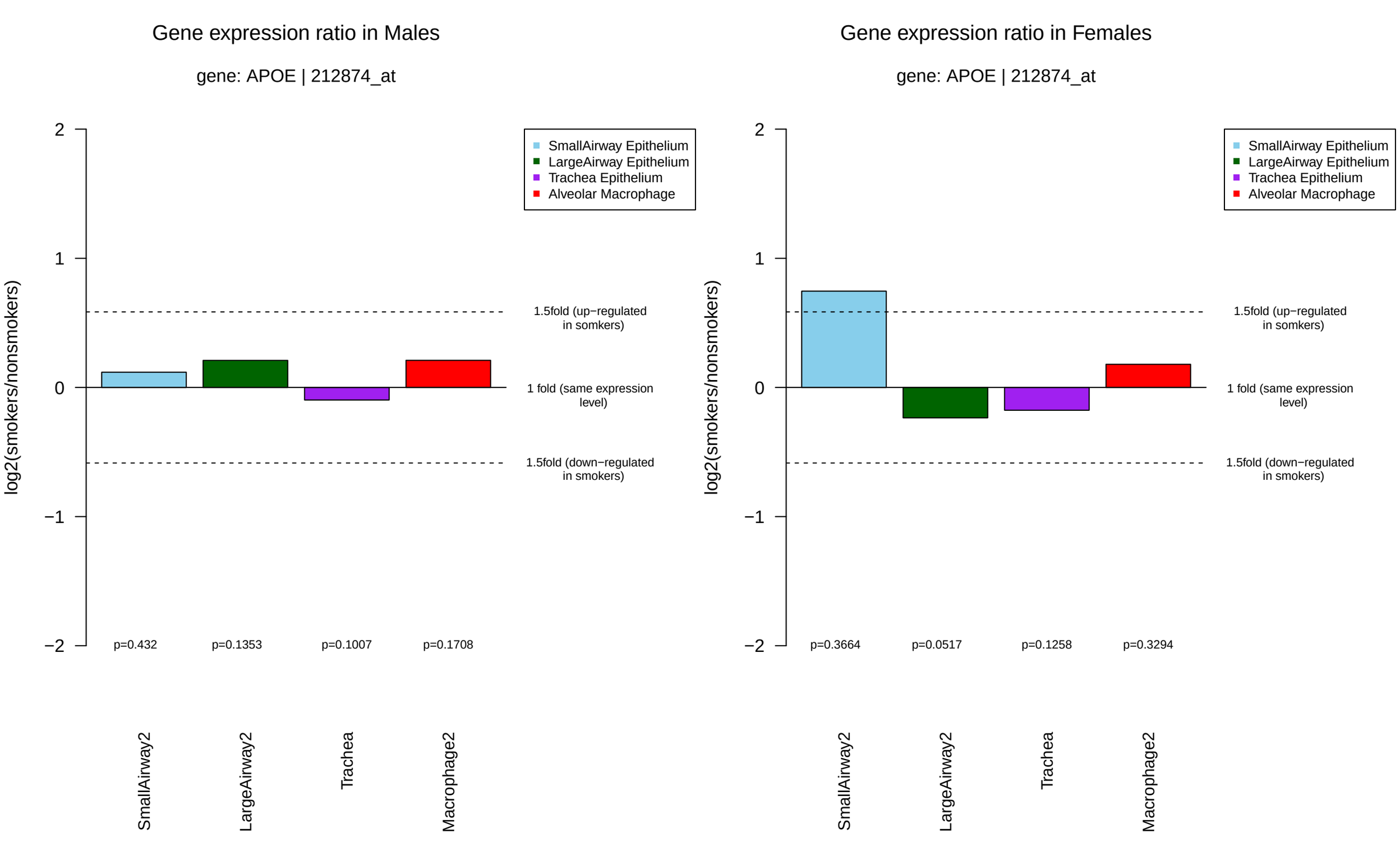
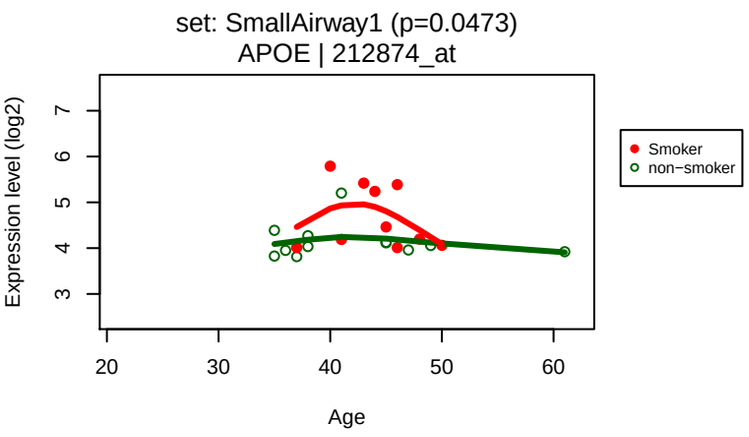
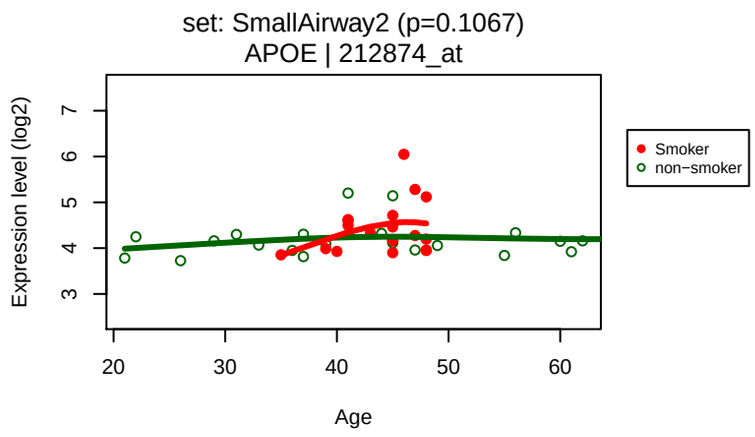
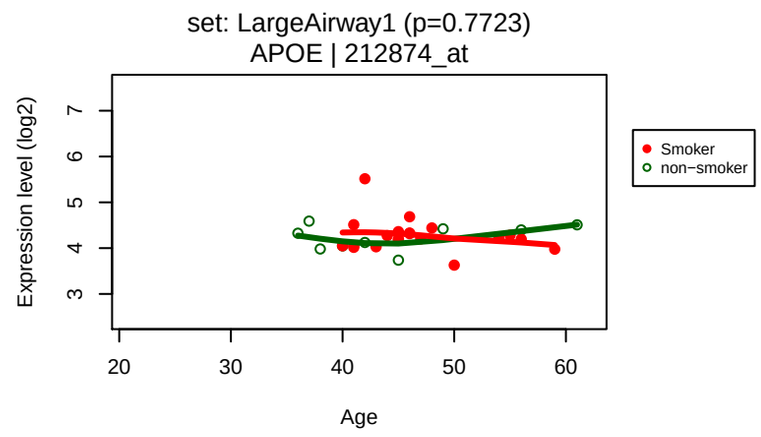

Gene expression of smokers and nonsmokers
gene: APOE | 212874_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0473
p=0.1067
p=0.1932
p=0.7723
p=0.5026
p=0.0372
p=0.579
p=0.083
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

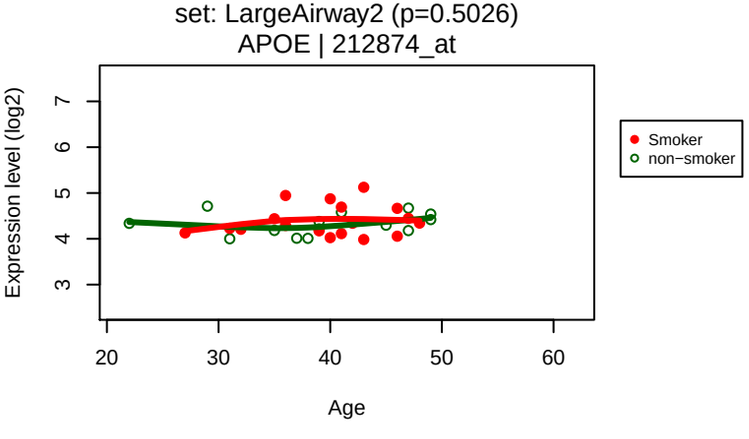

Gene expression ratio (smokers/nonsmokers)
gene: APOE | 212874_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0473
p=0.1067
p=0.1932
p=0.7723
p=0.5026
p=0.0372
p=0.579
p=0.083
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

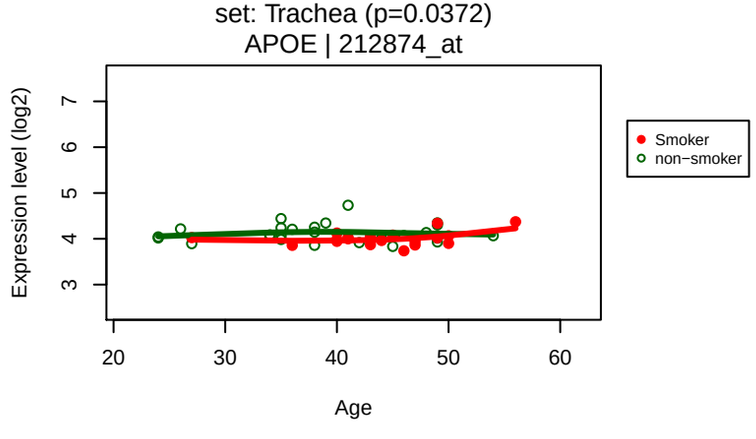

Gene expression ratio in Males
gene: APOE | 212874_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.432
p=0.1353
p=0.1007
p=0.1708
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: APOE | 212874_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3664
p=0.0517
p=0.1258
p=0.3294
SmallAirway2
LargeAirway2
Trachea
Macrophage2

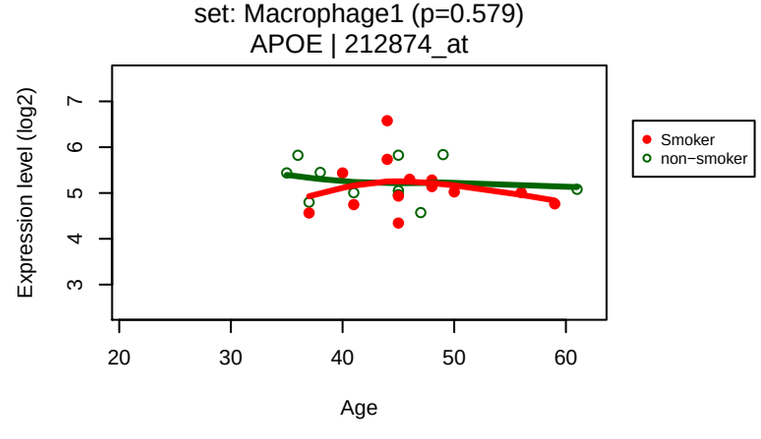

set: SmallAirway1 (p=0.0473)
APOE | 212874_at
Expression level (log2)
Age
Smoker
non−smoker

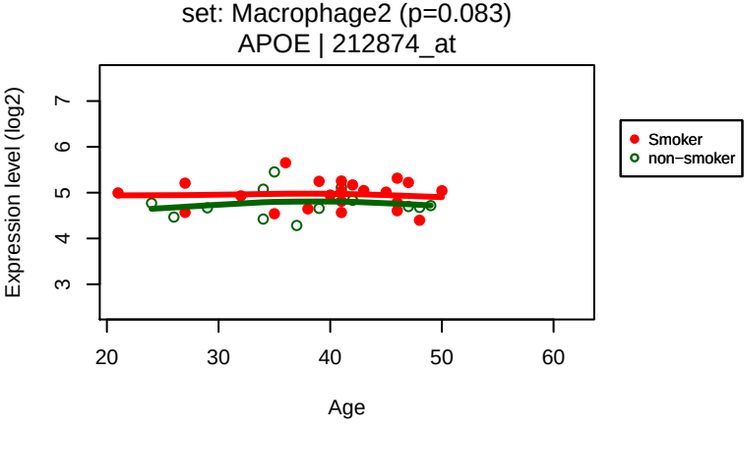

set: SmallAirway2 (p=0.1067)
APOE | 212874_at
Expression level (log2)
Age
Smoker
non−smoker

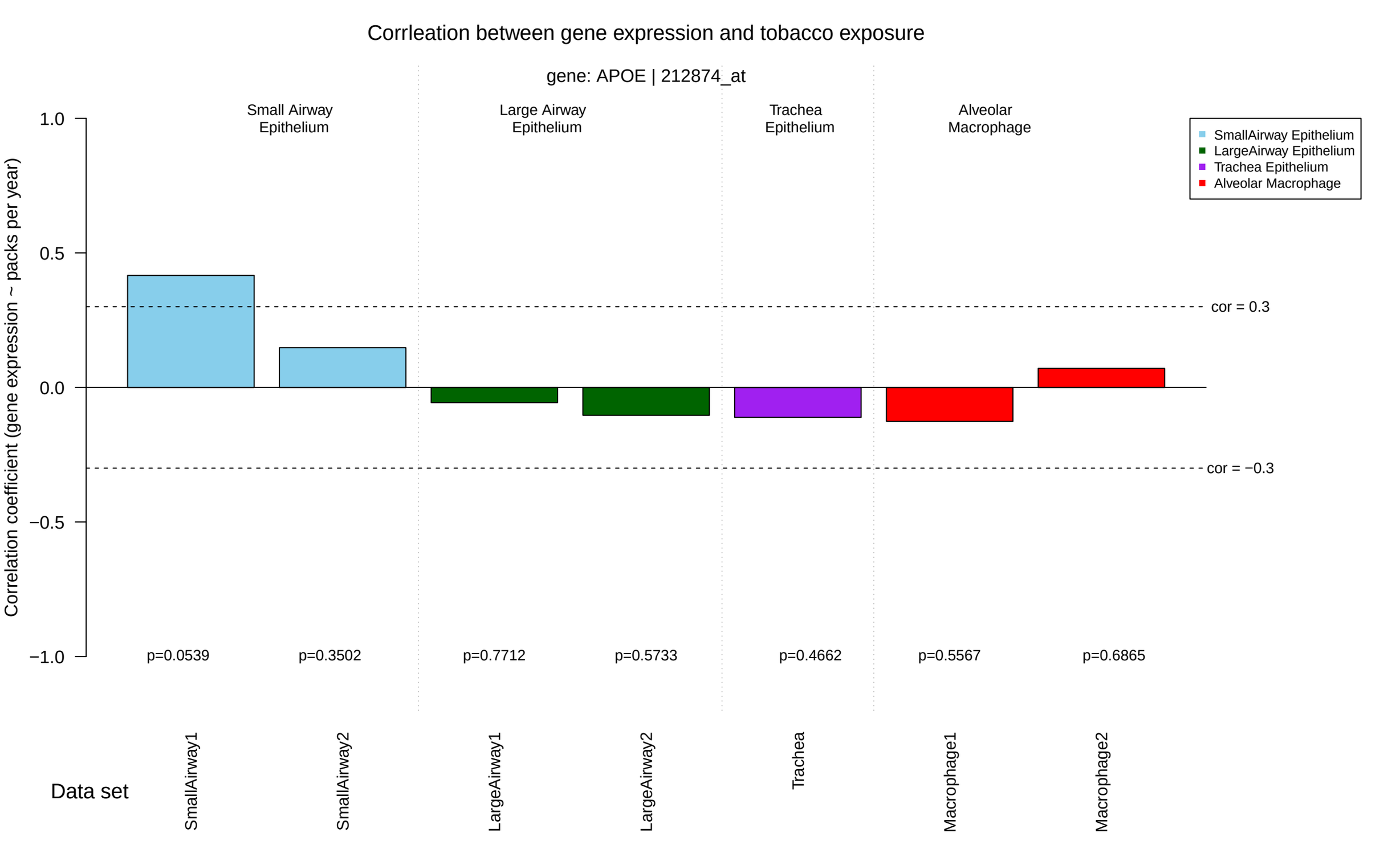

set: LargeAirway1 (p=0.7723)
APOE | 212874_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5026)
APOE | 212874_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0372)
APOE | 212874_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.579)
APOE | 212874_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.083)
APOE | 212874_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: APOE | 212874_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0539
p=0.3502
p=0.7712
p=0.5733
p=0.4662
p=0.5567
p=0.6865
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2